Friction Mini-Assessment

TOTAL:39

_____

1. **State** three everyday activities you couldn't do without friction.

_____
_____

_____
_____

(3 marks)

2. **List** three ways to reduce friction.

   a. _____
   _____

   b. _____
   _____

   c. _____
   _____

(3 marks)

3. **State** whether the following are true or false.

   a. Friction acts in the same direction as an object's motion.
   _____

   b. The greater the weight of the sliding object, the less the force of friction it experiences.
   _____

   c. Friction makes a machine more efficient.
   _____

   d. Streamlining reduces friction.
   _____ (4 marks)

4. Without friction, you could not pick up a basketball. **Explain** why.

_____
_____

_____
_____

_____
_____

(2 marks)

5. A mountain bike is designed to be considerably heavier than a road bike.

Explain why this is the case, considering the typical surface each is used on.

_____
_____

_____
_____

_____
_____

(2 marks)

6. **Calculate** the average friction force measured when moving each block.

Object Moving	Friction Force Measured			Average
	Trial 1	Trial 2	Trial 3	
Block	12	13	11	
Block + 100g	14	15	15	
Block + 200g	16	17	17	
Block + 300g	19	18	19	
Block + 400g	21	21	20	

(5 marks)

7. **Classify** each of these as increasing or decreasing friction.

a. Increasing the weight of an object.
_____

b. Making a surface rougher.
_____

c. Streamlining an object. _____

d. Using oil in an engine.
_____

e. Adding small bumps on a basketball surface.
_____

f. Using ball bearings in a wheel. _____

g. Polishing the bottom of skis. _____

(7 marks)

8. **Compare** the level of friction present between a new pair of running shoes and a very old worn pair of running shoes.

_____
_____

_____
_____

_____
_____

(2 marks)

9. **Explain** why this is the case.

_____
_____

_____
_____

_____
_____

(2 marks)

10. The minivan below is travelling along a flat road. Use the picture to answer the following questions.



a. **Name** two sources of friction that are acting on the minivan.

_____
_____

(2 marks)

b. **State** the direction in which these forces act.

_____
_____

(2 marks)

c. The surfboard is now removed from the roof-rack.

i) **State** whether the friction acting on the car would increase or decrease.

_____
_____

_____
_____

_____
_____

(1 marks)

ii) **Justify** your answer.

_____
_____

_____
_____

_____
_____

(2 marks)

11. Weightlifters rub chalk into their hands before attempting a lift. The chalk absorbs any sweat on the weightlifter's palms.

Propose why weightlifters use this chalk.

_____
_____

_____
_____

_____
_____

(2 marks)